

ALTDORF, Switzerland

WMO: 066720

Lat: 46.8869N

Lon: 8.6217E

Elev: 1480

StdP: 13.93

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.9	23.0	11.7	10.5	25.9	15.2	12.5	27.6	35.4	52.1	31.5	51.4	4.4	140	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	84.2	67.3	81.1	66.4	78.2	65.3	69.6	81.0	68.0	78.5	66.6	76.1	10.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	99.4	75.5	64.0	94.4	73.6	62.8	90.3	72.2	34.5	81.0	33.1	78.7	32.0	76.4	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.1	24.0	18.7	14.9	89.5	4.2	3.2	11.8	91.8	9.4	93.7	7.0	95.5	3.9	97.8		
			14.0	72.8	4.0	1.7	11.1	74.0	8.7	75.0	6.4	75.9	3.5	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	35.1	36.8	43.4	50.5	58.0	64.0	66.8	66.0	59.0	52.1	42.8	36.5
	DBStd	13.11	7.04	7.09	6.78	7.14	6.62	6.58	5.47	5.38	5.56	6.31	7.48	6.93
	HDD50	1859	464	373	221	79	10	1	0	0	3	48	238	422
	HDD65	5391	927	790	669	437	233	97	44	52	192	402	665	883
	CDD50	2227	2	3	18	95	258	421	521	495	273	114	23	4
	CDD65	281	0	0	0	2	16	67	100	82	12	2	0	0
	CDH74	1825	0	0	0	6	121	489	644	502	56	7	0	0
	CDH80	383	0	0	0	0	14	106	146	114	2	1	0	0
Wind	WSAvg	5.4	5.4	5.8	6.0	7.2	6.2	5.1	4.8	4.2	4.4	5.2	5.4	5.3
Precipitation	PrecAvg	66.0	3.7	4.1	4.4	4.8	6.8	7.2	7.3	7.5	5.7	4.4	5.3	4.6
	PrecMax	83.1	8.7	12.6	9.6	9.9	12.2	11.7	13.9	15.1	9.8	8.4	16.4	11.7
	PrecMin	49.1	0.2	1.3	1.7	0.9	1.8	3.6	3.0	2.1	2.2	0.2	0.4	0.1
	PrecStd	7.6	2.3	2.6	2.0	2.0	3.0	2.0	2.6	2.8	2.0	2.0	3.1	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	60.3	65.9	74.6	82.2	87.7	87.9	88.0	79.1	74.2	67.3	62.7
		MCWB	45.7	45.2	50.5	56.3	62.7	68.2	69.3	69.1	63.3	56.5	51.5	47.8
	2%	DB	54.3	55.5	61.8	70.2	77.2	83.1	84.0	82.9	75.3	70.2	64.4	56.6
		MCWB	42.1	43.2	48.7	54.1	60.4	67.0	67.9	67.6	62.6	55.3	49.8	43.9
	5%	DB	49.2	51.5	58.6	66.8	73.9	79.5	80.6	79.3	72.0	65.6	59.3	49.9
		MCWB	42.3	41.6	47.2	52.3	58.7	65.5	67.1	67.1	61.2	55.1	47.4	43.4
	10%	DB	44.9	47.3	55.3	63.6	70.4	76.2	77.4	76.1	68.8	62.1	53.1	45.9
		MCWB	40.8	41.0	46.0	50.7	57.0	64.0	65.9	65.8	60.6	54.7	48.2	42.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.2	47.1	52.7	58.5	65.3	71.0	71.8	71.4	66.7	60.3	53.0	48.3
		MCDB	53.9	54.5	62.1	71.6	78.3	83.5	84.5	82.9	75.0	65.7	61.3	59.0
	2%	WB	44.7	44.8	50.2	55.6	62.3	68.5	69.5	69.4	64.6	58.5	51.1	46.3
		MCDB	50.4	52.6	59.1	67.2	73.1	80.3	80.9	80.0	72.3	63.9	59.1	52.5
	5%	WB	42.6	43.0	48.5	53.7	60.1	66.7	68.0	67.6	62.8	57.1	49.8	44.3
		MCDB	47.9	49.7	56.4	64.2	70.5	77.4	78.4	77.3	69.4	63.2	57.5	49.0
	10%	WB	40.8	41.2	46.7	51.8	58.3	64.9	66.5	66.2	61.2	55.5	48.1	42.2
		MCDB	45.0	46.8	53.7	61.2	67.9	74.4	75.8	74.7	67.2	61.2	53.6	46.0
Mean Daily Temperature Range	5% DB	MDBR	10.2	12.0	15.1	16.6	17.0	16.6	16.2	15.6	14.4	13.3	11.0	9.6
		MCDBR	16.2	18.5	20.9	21.9	23.3	22.8	22.0	21.3	19.5	18.1	17.4	15.5
		MCWBR	8.9	10.2	11.6	11.4	11.3	11.3	10.7	10.5	10.1	8.7	7.7	8.3
	5% WB	MCDBR	13.6	15.8	19.1	21.4	21.5	21.2	20.0	19.7	17.6	15.2	14.3	13.1
MCWBR		8.8	9.8	11.3	12.1	11.2	11.1	10.3	10.3	10.0	8.7	8.1	8.2	
Clear-Sky Solar Irradiance	taub		0.297	0.331	0.390	0.445	0.467	0.475	0.475	0.460	0.420	0.382	0.332	0.293
	taud		2.284	2.185	2.088	2.022	2.046	2.088	2.098	2.151	2.234	2.299	2.347	2.317
	Ebn at Noon		248	260	259	256	255	253	251	249	248	239	232	235
	Edh at Noon		24	34	43	51	51	50	49	44	37	30	23	21
All-Sky Solar Radiation	RadAvg		453	745	1130	1400	1541	1795	1770	1539	1182	769	459	376
	RadStd		54	91	159	217	155	163	157	146	118	88	64	51

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.92	N/A	N/A	N/A	N/A	
(47)	Regional (117 neighbors)	N/A	N/A	N/A	+1.30	+0.74	+0.58	N/A	N/A	+103	+27	

Nomenclature: See separate page